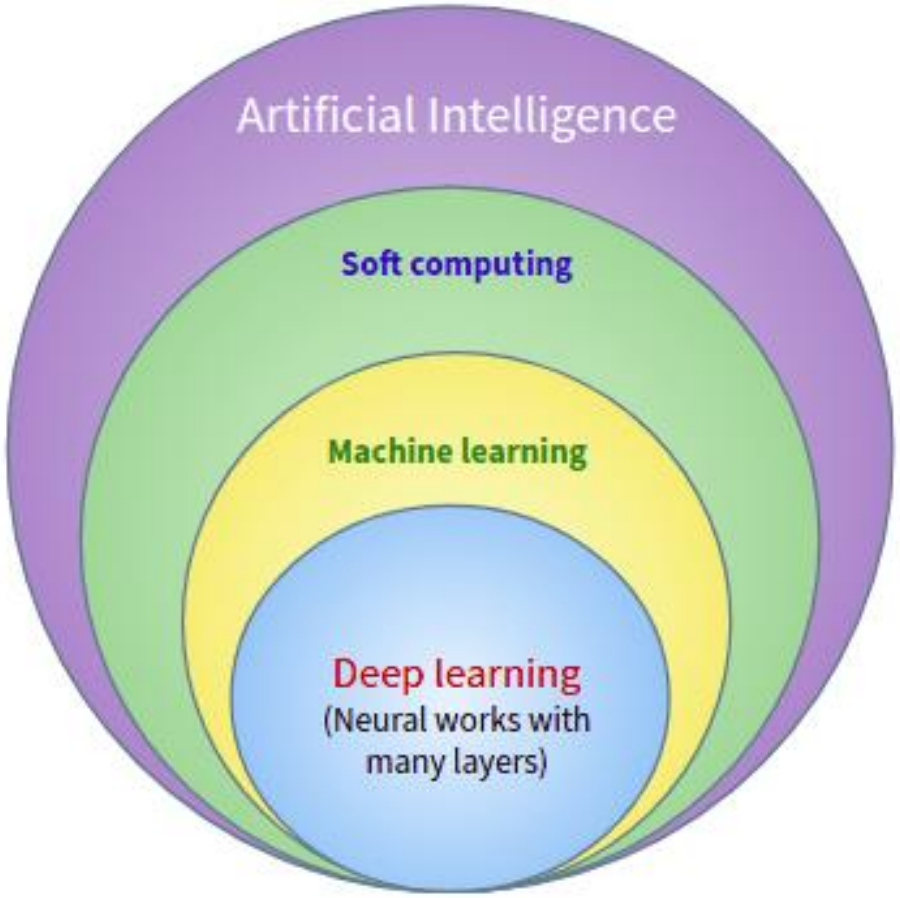
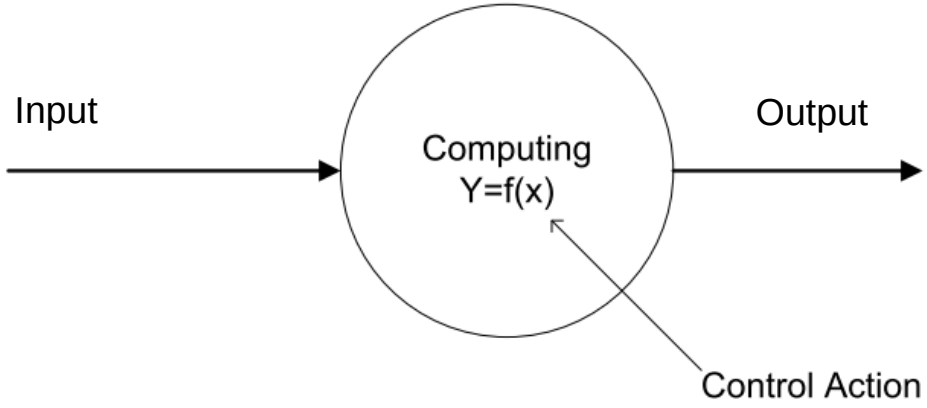
INTRODUCTION AND COURSE OUTLINE

Soft Computing – Lecture 1

Cognitive-based branches



Concept of computation



y=f(x), f is a mapping function (from x to y)

f is also called a formal method or an algorithm to solve a problem

Hard Computing (HC)

A specific computing is can be described as **hard computing** if:

- Precise result is guaranteed.
- Control action is unambiguous.
- Control action is formally defined with a mathematical model of an algorithm.

Examples of HC:

- Solving numerical problems (e.g., roots of polynomials, integration, etc.).
- Computational geometry problems (finding closest pair of points given a set of points).

Definition of Soft Computing (SC)

The term soft computing was proposed by the inventor of fuzzy logic, Lotfi A. Zadeh. He described it as follows:

"Soft computing is a collection of methodologies that aim to exploit the **tolerance for imprecision** and **uncertainty** to achieve **tractability**, **robustness**, and **low solution cost**. Its principal constituents are **fuzzy logic**, **neurocomputing**, and **probabilistic reasoning**. The role model for soft computing is the **human mind**."

Tolerance for imprecision: the results may not be accurate.

Uncertainty: if a problem is solved for several times the results may differ in each time.

Tractability: SC model can adapt with the changes in the environment.

Robustness: SC model can deal with inputs in different cases e.g., rotation in images.

Low solution cost: the computation in SC is much faster than HC, therefor, SC can solve a problem in real-time.

Goals of SC?

- Developing intelligent machines to provide solutions to real-world problems which are not modeled or too difficult to model mathematically.
- Exploiting the tolerance for imprecision, uncertainty, and partial truth for achieving close resemblance with human like decision making.

Hard Computing (HC) vs. Soft Computing (SC)

HC	SC
Conventional computing requires a precisely stated analytical model	Soft computing is tolerant of imprecision
Based on binary logic and numerical analysis	Based on neural computing, fuzzy logic, and probabilistic reasoning
Often requires a lot of computation time for real-world problem.	Can solve some real-world problems in reasonably less time (suitable for real-time problems)
Not suited for real world problems for which ideal model is not present	Suitable for real world problems
It requires full truth	Can work with partial truth (noisy data)
It is precise and accurate	Imprecise

Unique Features of SC

Soft Computing is an approach for constructing systems which are:

- Computationally intelligent.
- Possess human like expertise in particular domain.
- Can adapt to the changing environment and can learn to do better.
- Can explain their decisions.

Applications of SC

- Person identification
- Hand written character recognition
- Weather forecasting
- Medical diagnosis
- Robotics
- Washing machines
- and many others...

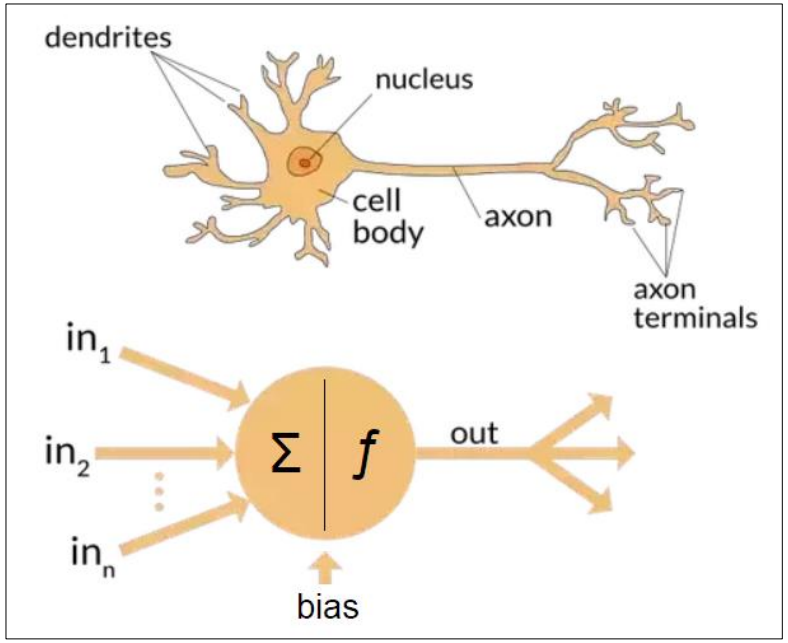
Components of SC

Components of soft computing include:

- Artificial Neural Network (ANN)
- Fuzzy Logic (FL)
- Evolutionary Computation (EC)

Artificial Neural Networks (ANN)

- Imitation of the natural intelligence of the brain.
- Parallel processing with incomplete information.
- Nerve cells function about 106 times slower than electronic circuit gates, but human brains process visual and auditory information much faster than modern computers.
- The brain is modeled as a continuous-time non linear dynamic system in connectionist architectures.
- Distributed representation in the form of weights between a massive set of interconnected neurons.



How soft computing in terms of ANN?

How a student learn form his teacher?

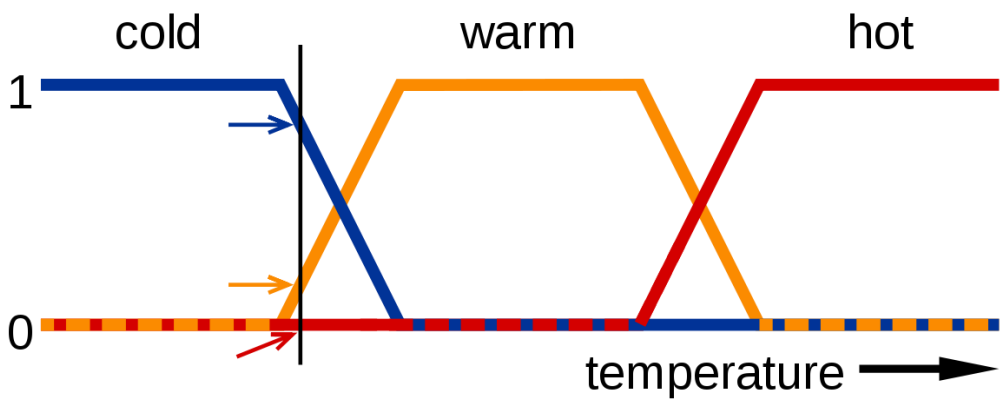
- Teacher asks questions then tell the answers.
- Teacher puts questions and hints answers then asks whether the answers correct or not.
- Student thus learn a topic and store in his/her memory
- Based on the acquired knowledge, he/she solves new problems.

This is an example of how a human brain works.

Based on this concept, ANN is used to solve problems.

Fuzzy Logic (FL)

- Human brains interpret imprecise and incomplete sensory information provided by perceptive organs.
- Fuzzy set theory provides a systematic calculus to deal with such information linguistically.
- It performs numerical computation by using linguistic labels stimulated by membership functions.



How soft computing in terms of FL?

How a doctor treat a patient?

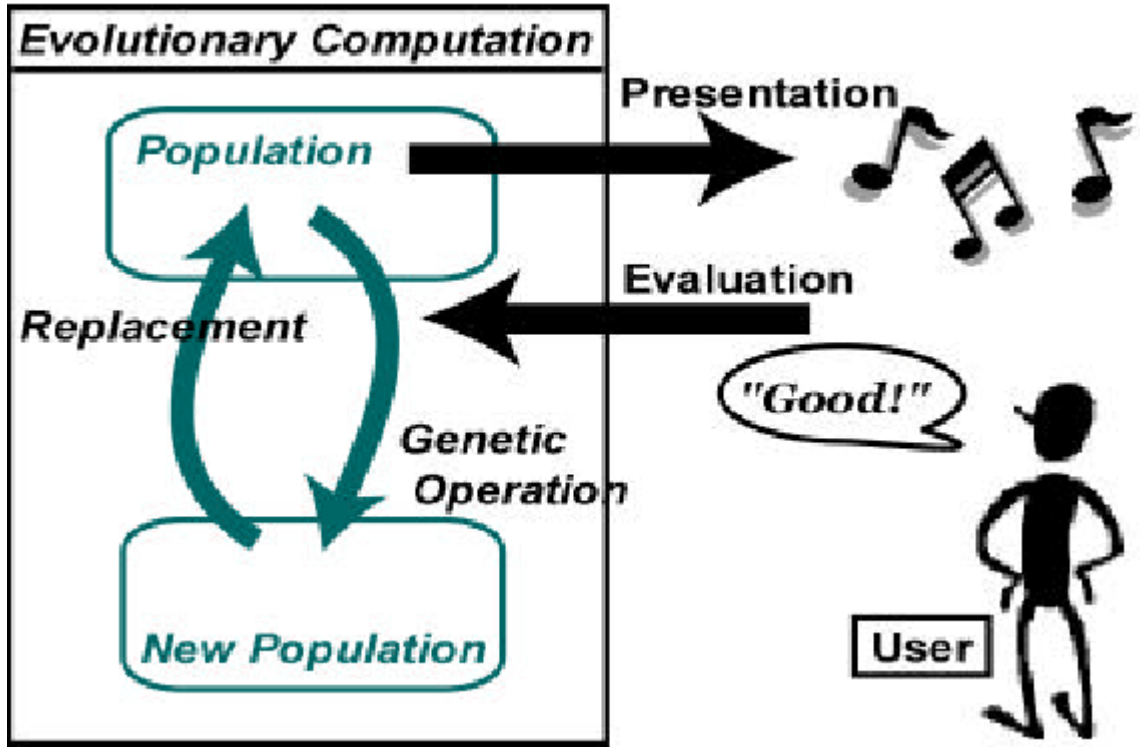
- Doctor asks the patient about suffering.
- Doctor finds the symptoms of disease.
- Doctor prescribes test and medicine.

Based on this concept, FL is used to solve problems.

- Symptoms are correlated with disease with uncertainty.
- Doctor prescribes tests/medicine fuzzily.

Evolutionary Computation (EC)

- Natural intelligence is the product of millions of years of biological evolution.
- Simulation of complex biological evolutionary processes.
- Genetic algorithm is one computing technique that uses an evolution based on natural selection.



How soft computing in terms of EC?

How someone selects the best?

- It starts with a population randomly.
- Reproduces another population (next generation).
- Rank the population and selects the superior individuals.

Based on this concept, EC is used to solve problems.

- This is a natural algorithm in life.
- Can be used for solving optimization problems.

Course outline

- Artificial neural networks (ANN)
  - Introduction to ANN
  - The perceptron
  - Multilayer networks
  - Training of NN
  - Backpropagation algorithm
- Fuzzy Logic (FL)
  - Introduction to FL
  - Fuzzy sets and membership functions
  - Fuzzy relations and rules
  - Fuzzification and Defuzzification
- Evolutionary computation (EC)
  - Introduction to EC
  - Genetic Algorithm (GA)
  - Evolutionary Algorithm (EA)
  - Genetic Programming (GP)
  - Simulated Annealing (SA)
  - Ant Colony Optimization (ACO)
  - Particle Swarm Optimization (PSO)
  - Bees Colony Optimization (BCO)